

ST512-SP26-Homework-5

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Problem 1

a

Source	df	SS	MS	F-value	p-value
Factor A	4	375	93.75	27.79	3.66e-14
Factor B	2	462	231.0	68.49	0
AxB Interaction	8	38	4.75	1.40	0.210
Error	75	253	3.373	–	–
Total	89	1128	–	–	–

b

Based on the ANOVA table, the interaction factor has a p-value of 0.210. Using a 0.05 significance level, we fail to reject the null hypothesis. Therefore, there is NOT a significant interaction between factor A and factor B.

c

Based on the ANOVA table, for Factor A the p-value was calculated to be 3.66e-14. Using a significance value of 0.05, we reject the null hypothesis. Therefore, there is a significant difference in mean response across the levels of factor A.

d

Based on the ANOVA table, for Factor B the p-value was calculated to be 0. Using a significance value of 0.05, we reject the null hypothesis. Therefore, there is a significant difference in mean response across the levels of factor B.

Problem 2

a

a=4, b=3, n=3, N=36, SSE=720

df (error)= 24, MSE=720/24=30.0

No interaction → each factor A mean averages over b*n=9

Tukey MSD:

$$q_{\alpha,4,24} * \sqrt{MSE/n_A} = 3.90 * \sqrt{30/9} = 7.12$$

Bonferroni MSD:

$$t_{\frac{\alpha}{2C},24} * \sqrt{2MSE/n_A} = 2.875 * \sqrt{60/9} = 7.42$$

b

Interaction → compare Factor A within one level of Factor B, n=3

Tukey MSD:

$$q_{\alpha,4,24} * \sqrt{MSE/n_A} = 3.90 * \sqrt{30/3} = 12.33$$

Bonferroni MSD:

$$t_{\frac{\alpha}{2C},24} * \sqrt{2MSE/n_A} = 2.875 * \sqrt{60/3} = 12.86$$

Problem 3

R code

problem 1

```
1-pf(27.79,4,75)
```

```
[1] 3.663736e-14
```

```
1-pf(68.49, 2,75)
```

```
[1] 0
```

```
1-pf(1.40,8,75)
```

```
[1] 0.2105261
```

problem 2

```
qtukey(.95,4,24)
```

```
[1] 3.901262
```

```
qt(0.0041667,24)
```

```
[1] -2.875091
```